

HapGUI

Haplotype analysis for rice and other crops
Desktop application for Windows

User Manual

Powered by IRRI South Asia Hub
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Version 0.1.0 · Runs offline on your computer. No R, Java or Docker installation required.

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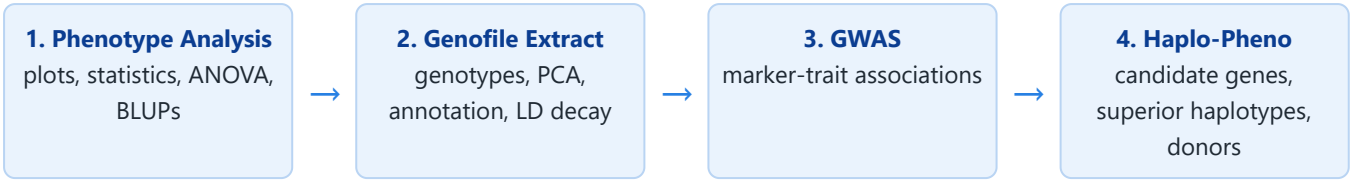
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1. About HapGUI

HapGUI takes you from **field phenotypes** to **superior haplotypes and donor lines** in one application. You do not need to write code. It works with two kinds of germplasm:

- **3K-RICE** – the built-in 3,000 Rice Genomes data (about 3,000 lines, 52,600 markers). You only supply phenotypes.
- **External** – your own germplasm. You supply phenotypes, a genotype file in VCF format and a gene annotation (GFF3) for the same genome.

The analysis workflow



Steps 2 to 4 depend on each other: **Genofile Extract must be run before GWAS, GWASpoly and Haplo-Pheno**, because it creates the marker table, the principal components and the per-gene SNP files that those steps use. The **Genotype File Converter** and the **Visualization** tab work on their own.

2. Installing and starting the app

System requirements

Item	Requirement
Operating system	Windows 10 or Windows 11, 64-bit
Memory	8 GB RAM recommended. Large external datasets (hundreds of thousands of markers or many samples) need more; close other programs while running Genofile Extract.
Disk space	About 3 GB for the program, plus working space for your results (a large VCF can produce several hundred MB of output).
Other software	None. R, Java, PLINK 2, TASSEL and the 3K-RICE data are all included in the installer.

Installation

1. Double-click `HapGUI Setup 0.1.0.exe`.
2. If Windows SmartScreen shows “Windows protected your PC”, click **More info** and then **Run anyway**. (The installer is not code-signed yet.)
3. Choose the installation folder (or keep the default) and click **Install**.
4. Start **HapGUI** from the Start menu or the desktop shortcut.

The first start takes 10 to 30 seconds while the analysis engine starts. When the badge at the top right reads **R backend connected**, the app is ready.

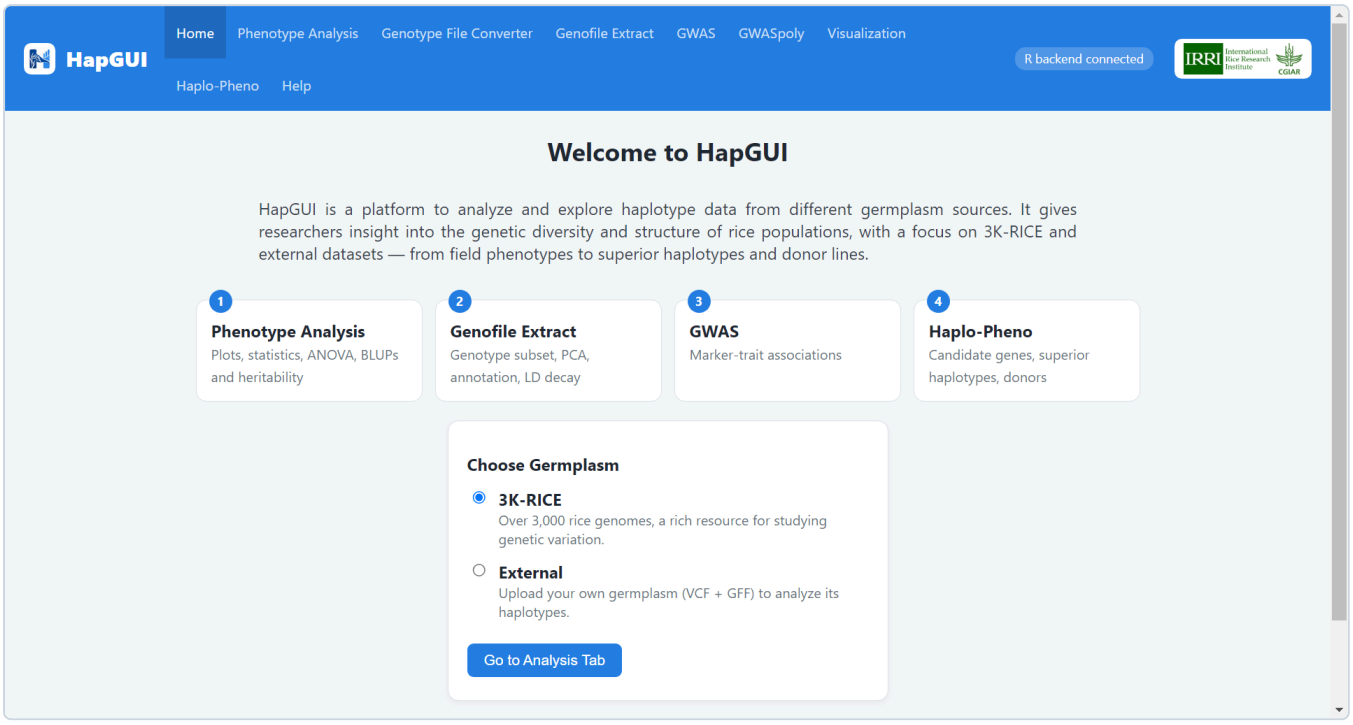


Figure 1. The Home screen. The green badge at the top right shows the analysis engine is running; the chips at the bottom confirm PLINK2, Java, TASSEL and the 3K-RICE data were found.

Uninstalling

Use **Settings** → **Apps** → **Installed apps** → **HapGUI** → **Uninstall**. Your results are kept in `%APPDATA%\hapgui-desktop` and can be deleted manually.

3. Quick start with the demo data

The installation includes demo files in the `input` folder inside the HapGUI installation folder (`…\HapGUI\resources\input`). This 10-minute tour runs the whole 3K-RICE workflow.

#	Tab	What to do
1	Home	Keep **3K-RICE** selected.
2	Phenotype Analysis	Upload `demo_3Krice_step1.csv` , click **Load file**, choose Location and Season as environments, click **Generate plots**. Explore the tabs. In **BLUP**, select the trait `PL` and click **Generate BLUPs**.
3	Genofile Extract	Choose `demo_3Krice_step2.csv` and click **Run extract** (about 3 minutes).
4	GWAS	Choose `demo_3Krice_step2.csv` (or click **Use BLUP output**), keep mrMLM selected, click **Run GWAS**.
5	Haplo-Pheno	Phenotype file: `demo_3Krice_step2.csv` . MTA file: click **Use GWAS output**. Keep LD region 25000 and **High**, click **Run Haplo-Pheno**.

The files for the **External** workflow are `demo_external_step1.csv` and `demo_external_step2.csv` (a millet example with a GFF3 file is in `input\others (millets)`).

4. Preparing your input files

All tables are **CSV files** (comma-separated, first row = column names). In Excel use *Save As → CSV (Comma delimited)*.

4.1 Phenotype file for "Phenotype Analysis"

One row per observation. It needs a genotype/accession column and one or more numeric trait columns. Environment (e.g. location, season) and experimental-unit (e.g. block) columns are optional but enable the environment-wise summaries, correlations, ANOVA and BLUPs.

Location	Season	BLOCK	Accession	NPT	PL	GYKGPHA
ISARC	20WS	1	POHHERLIMASION::IRGC 62025-1	5	28.9	2011.7
ISARC	21WS	1	NAN TEO 14::IRGC 7304-1	10	25.4	3512.1

- Missing trait values may be left empty.
- **3K-RICE:** accession names must match the 3K list. You can identify lines by **Accession** name, **IRIS_ID** (e.g. `IRIS_313-11586`) or **IRGC_NO** (e.g. `IRGC 62025-1`); pick which one your file uses in the app.
- **External:** any names; optionally a column giving a group/category (e.g. subpopulation).

4.2 Phenotype file for Genofile Extract, GWAS and Haplo-Pheno

Any CSV with a column of line IDs and a numeric trait column. In Genofile Extract, GWAS, GWASpoly and Haplo-Pheno you choose the file and then pick the **ID column** and the **Trait column** in the app, so the file may contain other columns as well. This matters for the **BLUP output** of the Phenotype tab, which has one adjusted column per trait you selected plus a `subpopulation` column: choose the one trait you want to analyse (only numeric columns are offered).

X.Phenotype.	PL(adjusted)
IRIS_313-11586	27.23
IRIS_313-11037	27.83

> **IDs must match the genotype data.** For 3K-RICE use the IRIS IDs (`IRIS_313-…`). For external data the IDs must be exactly the sample names in your VCF. Lines that are not found are ignored; if none are found the run stops with an error.

4.3 External germplasm files

File	Requirement
VCF	`.vcf` or `.vcf.gz`, with sample names matching your phenotype IDs. Markers with minor allele frequency below 0.05 are removed automatically.

GFF3	Gene annotation of the **same genome version** as the VCF. Chromosome names do not have to be numbers. `1`, `chr1`, `Chr01` and Roman `I` are treated as the same chromosome, so a VCF using `1` works with a GFF3 using `Chr1`. Names such as `LG01` or `1A` also work; the analysis tools need numbers, so HapGUI codes them internally and shows your own names in every result.

ID mapping file (optional)	Use it when the sample names in the VCF are spelled differently from the IDs in your phenotype file (for example `IRIS-313-7690` in the VCF but `IRIS_313-7690` in the phenotype file). A plain text file with **two columns on each line: the ID used in the VCF, then the new ID** (separated by a tab, comma, semicolon or spaces). A header line and extra columns are ignored, and if the columns are the wrong way round HapGUI notices and swaps them. Samples that are not listed keep their own name.
	`IRIS-313-7690  IRIS_313-7690`
	`IRIS-313-7736  IRIS_313-7736`

Polyploid VCF	Genotypes with more than two alleles per call (for example `0/0/0/1` for a tetraploid) are detected automatically; see section 8.
Category (optional)	A column in the phenotype file that groups lines (e.g. subpopulation); it colours the PCA plot.

4.4 MTA file for Haplo-Pheno

A CSV listing marker-trait associations. HapGUI looks for a column whose name starts with “chr” and one that contains “pos”; if none are found it uses the first two columns. The GWAS tab produces a suitable file (`Identified_MTAs.csv`); you can also supply your own list.

Chromosome	Position
6	27688376
7	16888581

4.5 Files for the Visualization tab

Plot	File
Manhattan, SNP density	CSV with four columns: **Marker, Chrom, Position, −log10(p)**. The fourth column must already be on the −log10 scale. Any chromosome naming (1, chr1, 1A …) works.
Circular Manhattan	As above, with one additional −log10(p) column for each trait or model to draw as a ring.
LD Heatmap	A VCF file. The first 100 SNPs of each chromosome are used.

5. Home: choosing the germplasm

On the Home screen select **3K-RICE** or **External**. This choice is used by Phenotype Analysis (how lines are matched and grouped) and by Genofile Extract (which genotype data are used). You can change it at any time, but run the analysis steps for the same germplasm in order.

6. Phenotype Analysis

Loading data

1. Click **Browse…** and choose your phenotype CSV, then click **Load file**. The number of rows and columns is shown.
2. Set **Genotype column** (the accession/line column).
3. Optionally set **1st/2nd environment** and **1st/2nd experimental unit** columns.
4. 3K-RICE: choose the **Genotype ID type**. External: optionally choose a **Category column**.
5. Click **Generate plots**.

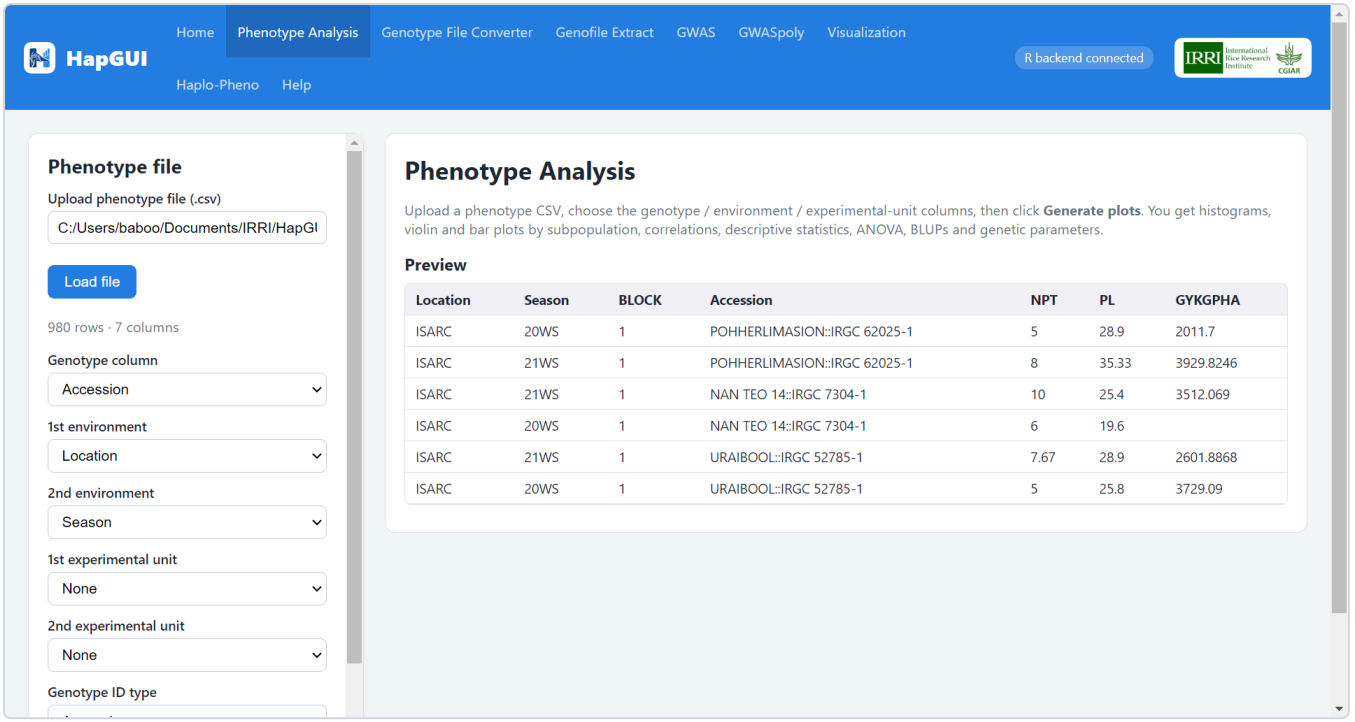


Figure 2. After loading the file, map the columns and click Generate plots.

The analysis tabs

Tab	What it does
Histogram	Distribution of a trait; optionally restricted to one environment/unit using the drop-downs.
Violin / Bar plot	Trait by subpopulation (3K-RICE groups: indica, aus, japonica, admix, aromatic) or by your category column. A second plot groups by an environment or unit column.
Correlation	Pairwise correlations between selected traits, and the correlation of one trait across environments. The plot grows with the number of columns; up to 16 environment / unit combinations can be drawn, so pick fewer environment or unit columns if you get a message about too many.
Descriptive statistics	Minimum, maximum, mean, standard deviation, skewness and kurtosis, overall and per environment/unit. Tables can be downloaded.
ANOVA	Augmented RCBD analysis of variance for a chosen treatment, block and trait.

BLUP	Best linear unbiased predictors (adjusted means) per genotype, from a model with genotype and block as random effects. Produces the file used by the later steps.
Genetic parameters	Genotypic and phenotypic variance, GCV, PCV, heritability and genetic advance.

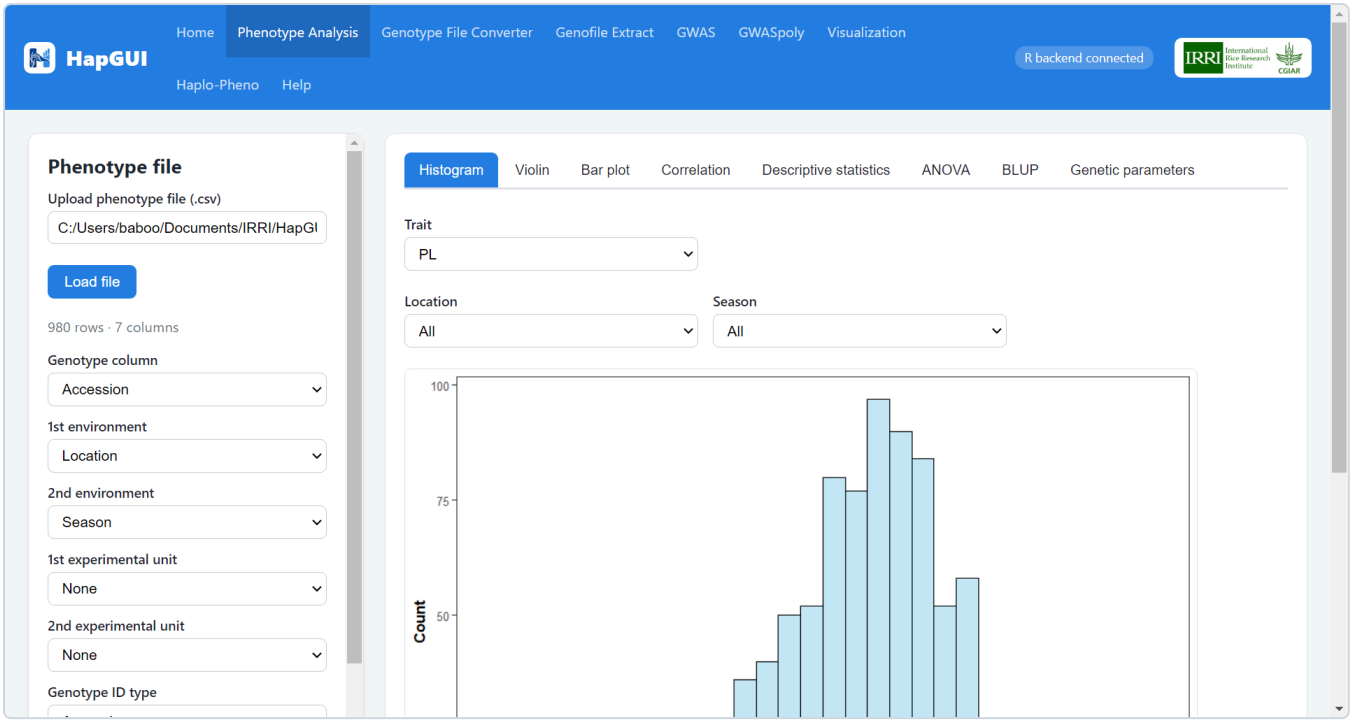


Figure 3. Histogram of the trait PL. Each plot has a download link beneath it.

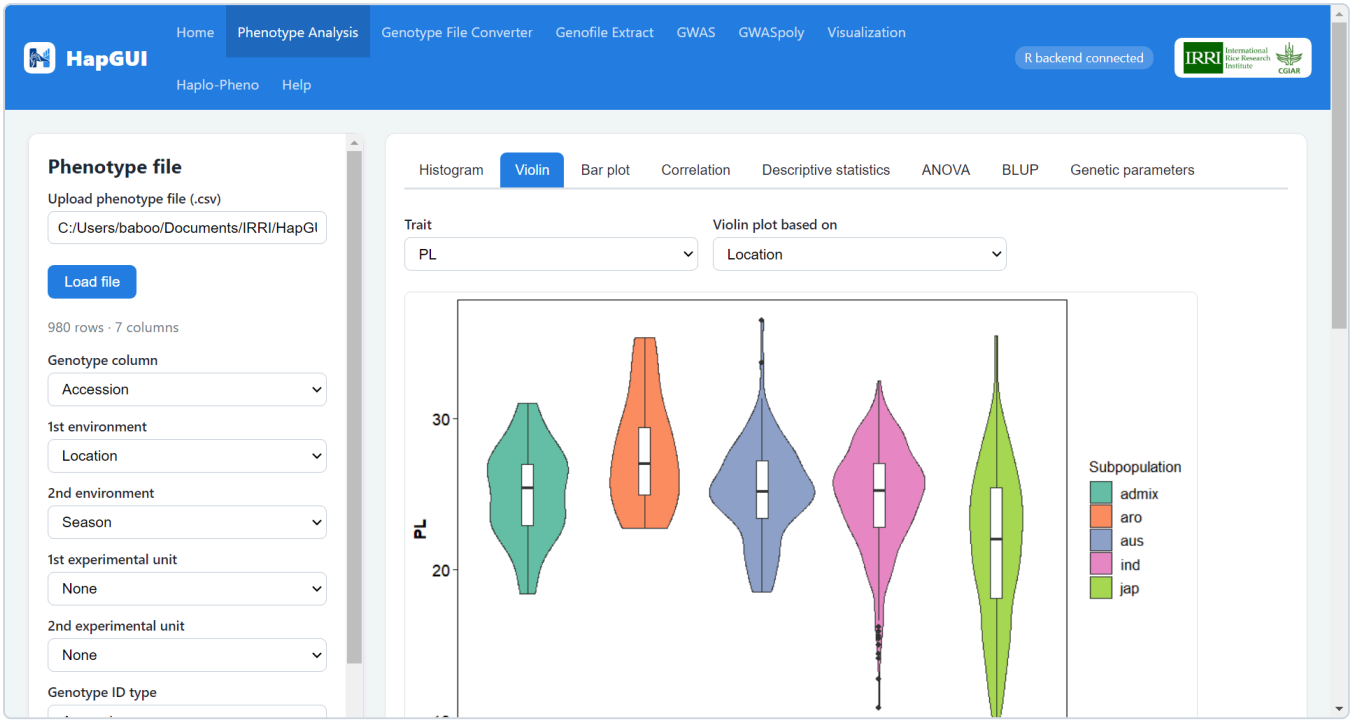


Figure 4. Violin plot of a trait by subpopulation.

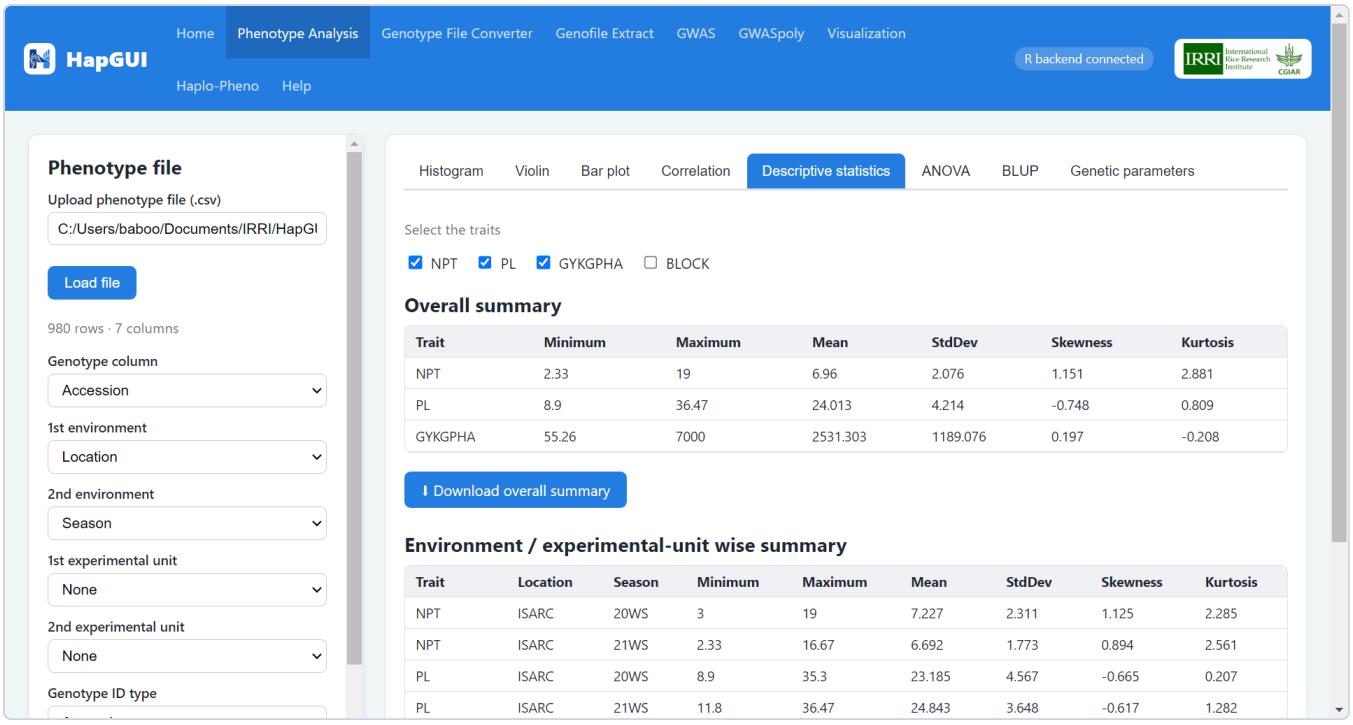


Figure 5. Descriptive statistics.

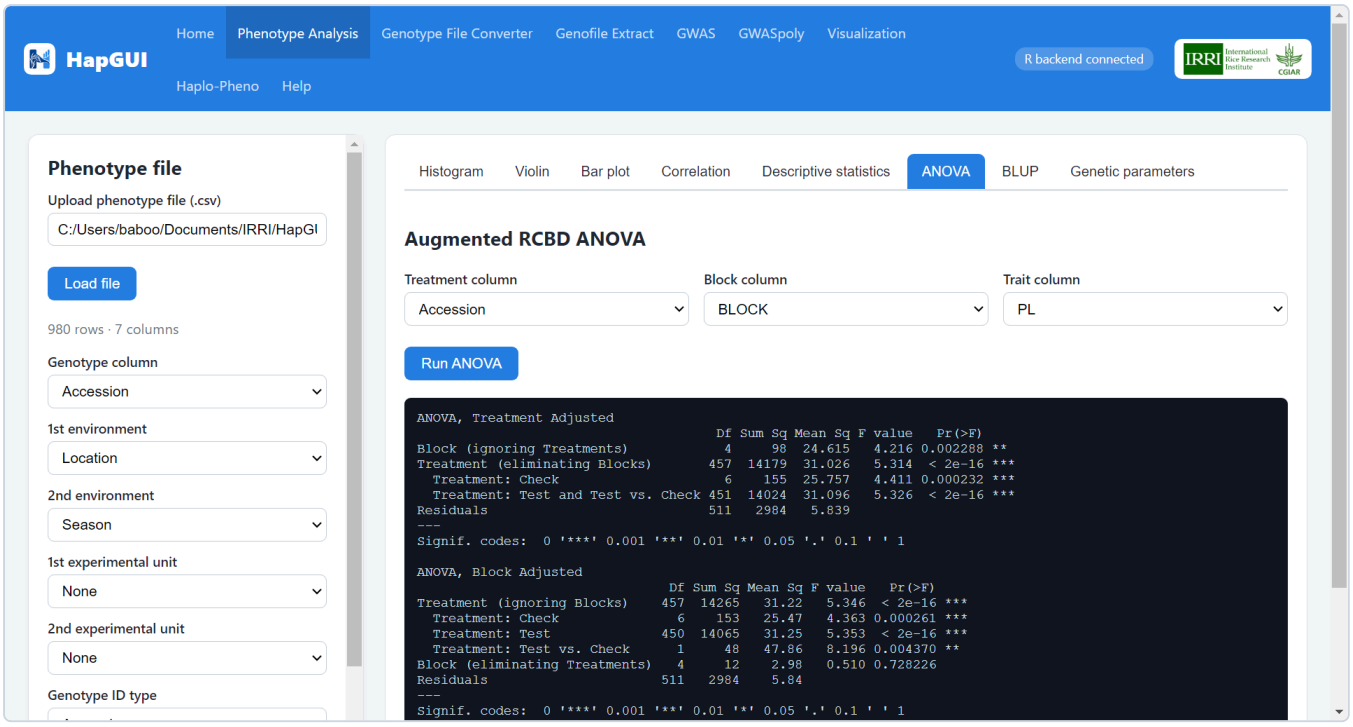


Figure 6. Augmented RCBD ANOVA output.

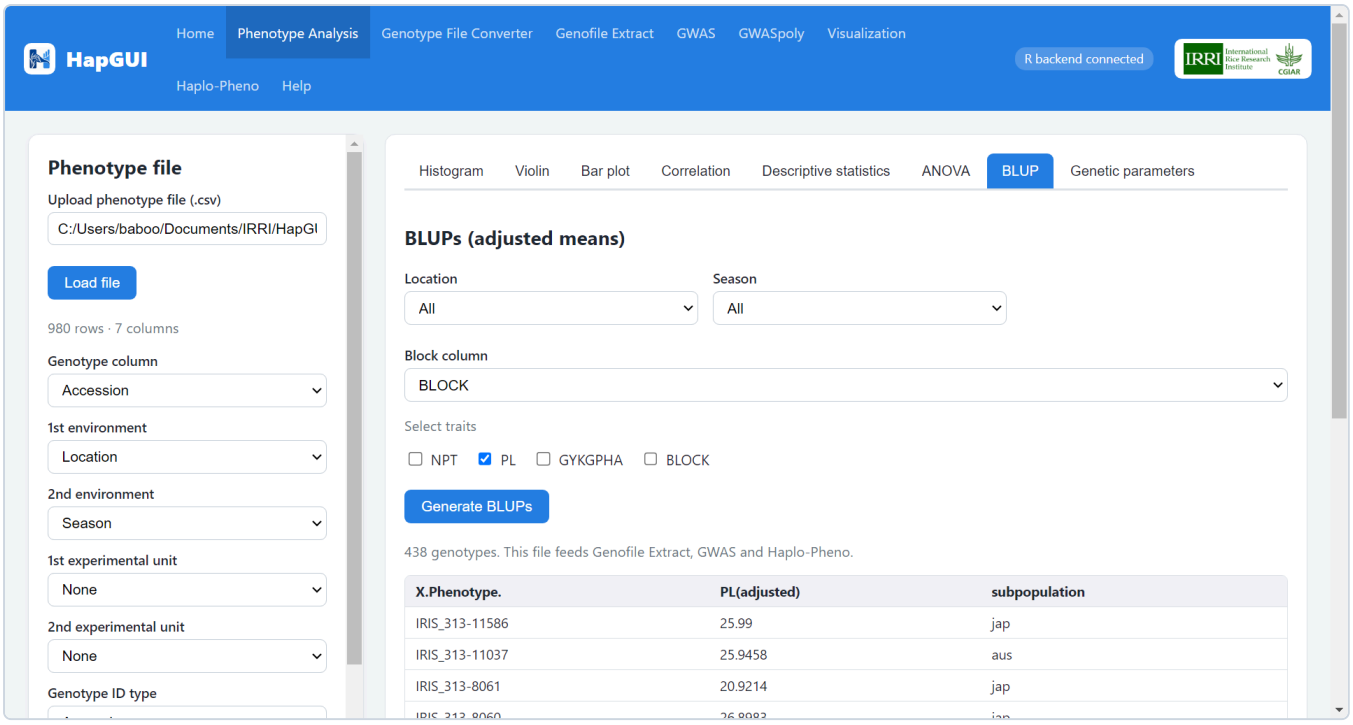


Figure 7. BLUPs. Download the file, or use the “Use BLUP output” buttons in the GWAS and Haplo-Pheno tabs.

Tip. Select only one trait in the BLUP tab if you plan to continue to GWAS: later steps analyse the first trait column of the file.

7. Genotype File Converter

Converts between common genotype formats. It is independent of the other tabs.

Conversion	Input
VCF to HapMap	one `.vcf` file
HapMap to VCF	one `.hmp.txt` file
PLINK to VCF	the three files `.bed`, `.bim`, `.fam`

Choose the conversion, select the file(s), and click **Run conversion**. The first lines of the result are previewed; click **Download converted file** to save it.

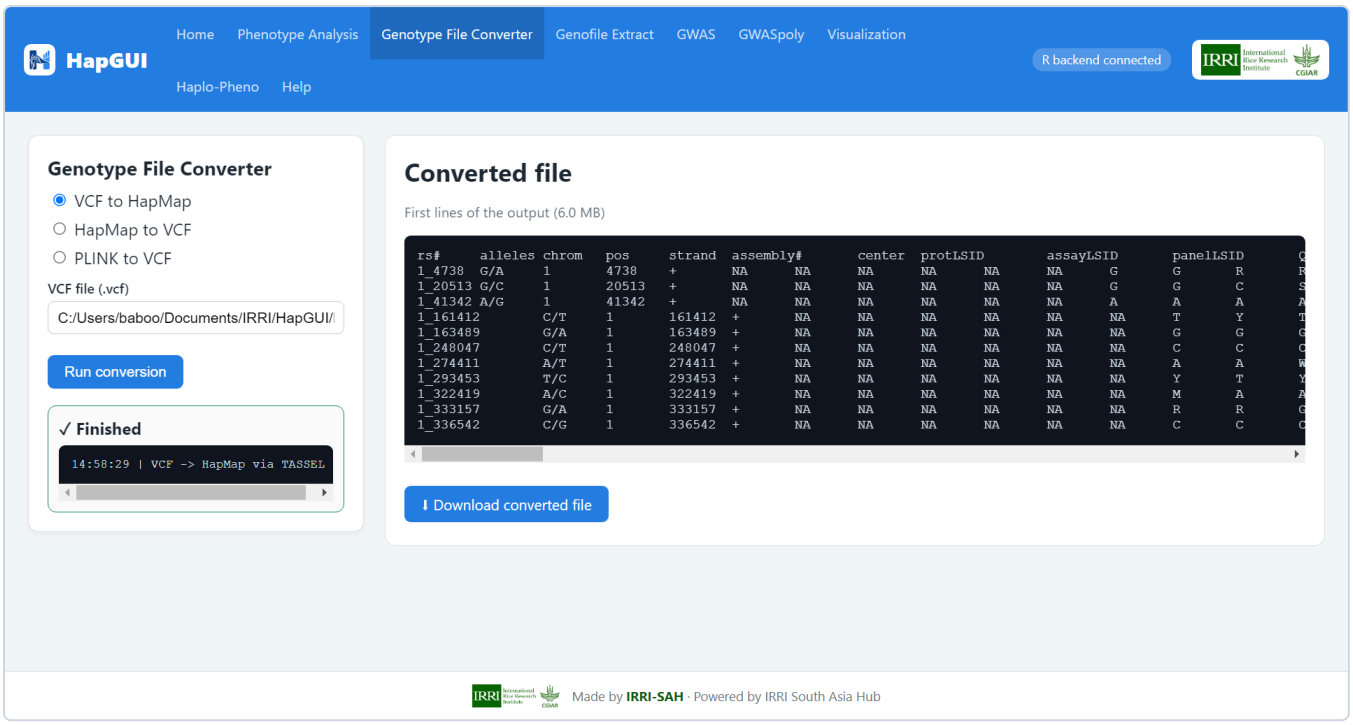


Figure 8. Converter result with preview.

8. Genofile Extract

Prepares everything the genetic analyses need from the lines in your phenotype file:

- selects those lines from the genotype data (3K-RICE built-in, or your VCF);
- converts to HapMap and a marker table;
- computes principal components (PCA) for population structure;
- annotates every variant (promoter, coding, intron, …) and the gene it falls in, and writes one SNP file per gene for Haplo-Pheno;
- estimates LD decay.

Running it

1. Choose the phenotype file, then the **ID column** (the lines to analyse) and the **Trait column**. The BLUP file from the Phenotype tab works; pick one trait.
2. **External only:** choose the VCF and the GFF3 file. Optionally choose an **ID mapping file** (section 4.3) and the category column.
3. Click **Run extract**. A progress box lists each step with a time stamp.

> **Sample IDs are checked first.** Before any work starts, HapGUI compares your phenotype IDs with the VCF's sample names (after applying the ID mapping file, if you gave one). If none match, it stops immediately and shows example IDs from both sides, and says if they would match when punctuation and capitals are ignored. If only some match, the analysis continues with those lines and the log reports how many.

Polyploid VCF files

Tools such as PLINK cannot read polyploid genotypes, so for a VCF whose calls have more than two alleles (`0/0/0/1` , ploidy 4, and so on) HapGUI reads the genotypes itself:

- Only biallelic SNPs with minor allele frequency ≥ 0.05 and at least half of the samples called are kept.
- Genotypes become allele dosages. `marker.csv` is written in GWASpoly's format (for example `GGGA` for three reference and one alternative allele), the PCA and the LD decay are computed from the dosages, and variants are annotated as usual.
- Use the **GWASpoly** tab for the association analysis; its ploidy is filled in from the extracted data. **GWAS (mrMLM) and Haplo-Pheno need diploid data** and tell you so if the last extraction was polyploid.

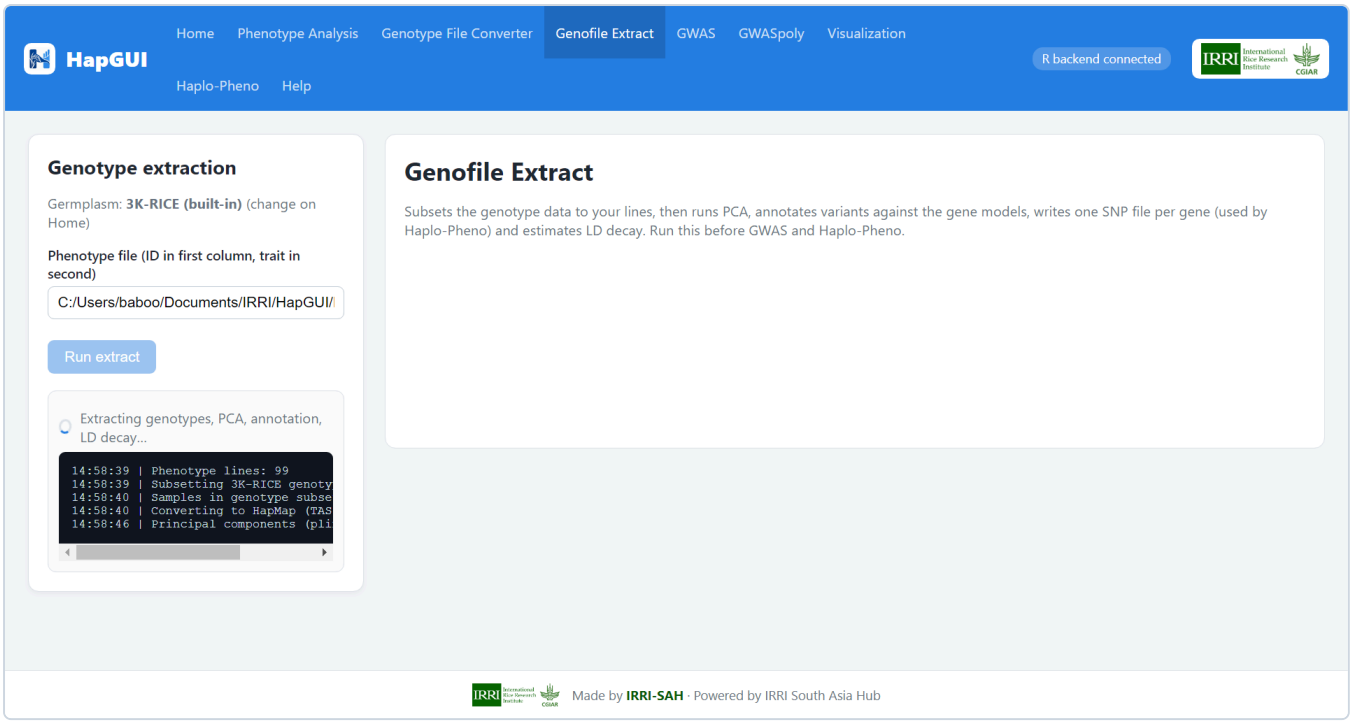


Figure 9. Progress while the extract runs.

Large VCF files. A VCF with hundreds of thousands of variants takes much longer (tens of minutes) and needs more memory; the progress box shows each stage (reading, classifying variants, writing per-gene files with a running count). Close other programs first, and prefer a VCF filtered to the lines you need.

How long does it take? The 3K demo (99 lines, 52,600 markers) takes about 2 to 6 minutes; most of this is the annotation step. Time grows roughly in proportion to the number of markers. You can switch to other tabs while it runs.

Results

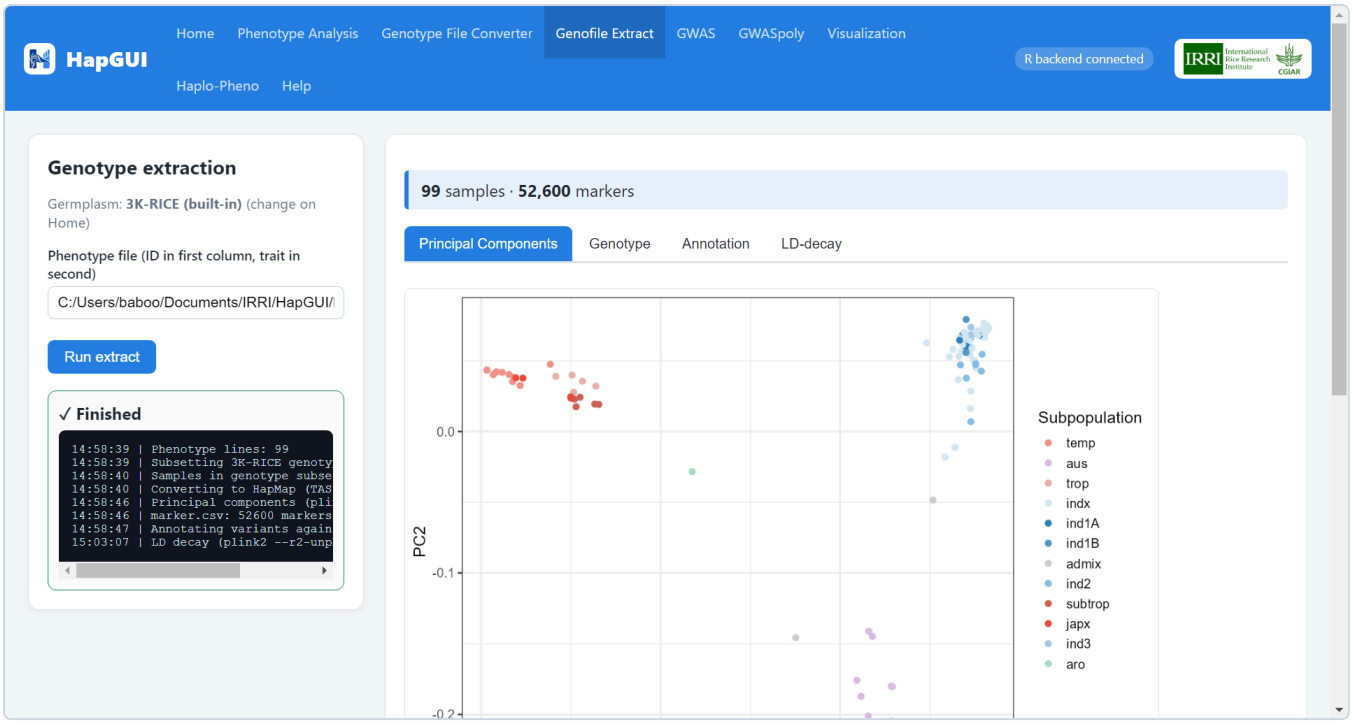


Figure 10. PCA coloured by subpopulation (3K-RICE) or by your category column (external).

Tab	Content
Principal Components	PC1 vs PC2 plot and the PCA table (downloadable).
Genotype	Preview of the marker × line table; the complete table can be downloaded.
Annotation	Each variant with its location class (coding, fiveUTR, threeUTR, exon, intron, promoter = within 2 kb upstream of a gene, or intergenic) and the gene it lies in or just upstream of. Where a variant falls in several transcripts, the most informative class is reported (coding before UTR before intron).
LD-decay	Mean r^2 against physical distance (up to 500 kb) with a fitted curve.

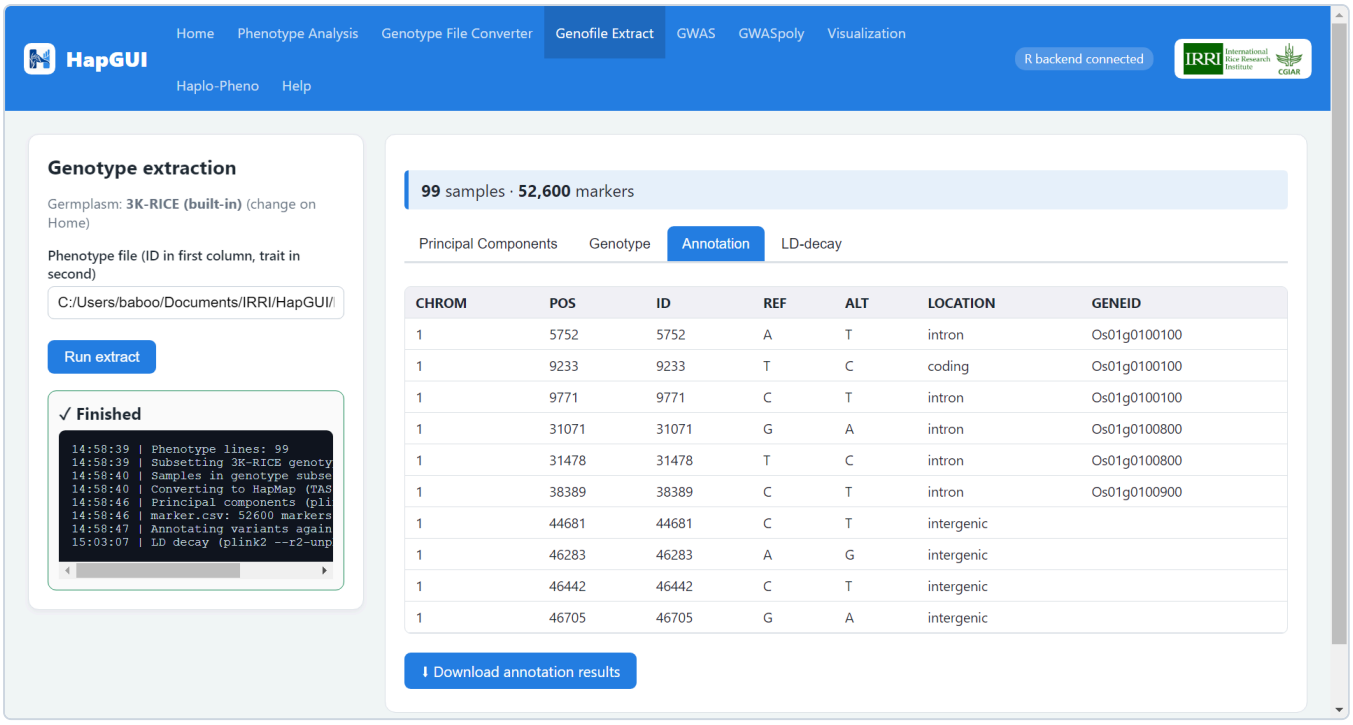


Figure 11. Variant annotation.

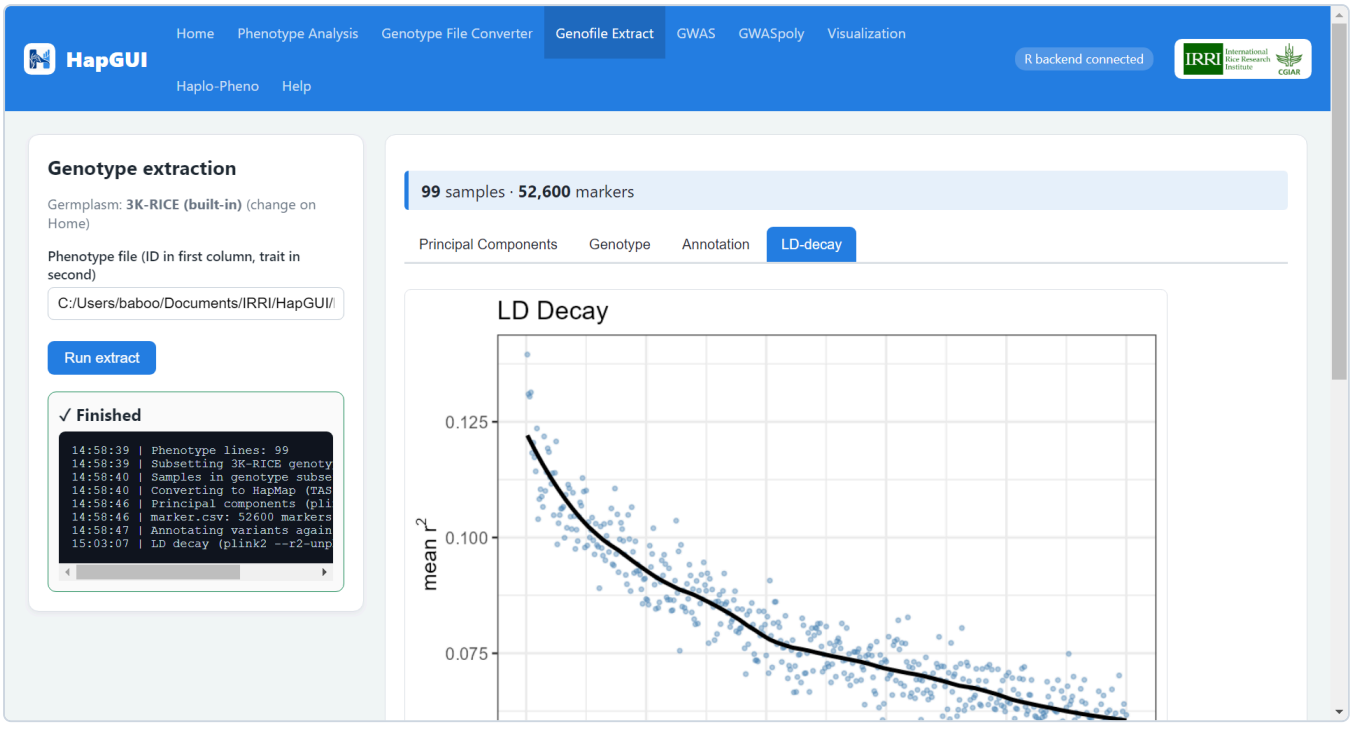


Figure 12. LD decay.

9. GWAS

Genome-wide association with the multi-locus mrMLM family of methods, using the marker table and principal components from Genofile Extract (so run Extract first, with the **same** phenotype lines).

Setting	Meaning
Phenotype file, ID column, Trait column	Any CSV with line IDs and a numeric trait. After choosing the file, pick the ID column and the one trait to analyse. **Use BLUP output** fills in the BLUP file from the Phenotype tab.
Principal components	Number of PCs used to correct for population structure (default 3, maximum 10).
Methods	mrMLM, FASTmrMLM, FASTmrEMMA, ISIS EM-BLASSO, pLARmEB. Selecting several methods takes longer; a marker found by several methods is listed once with all methods named.

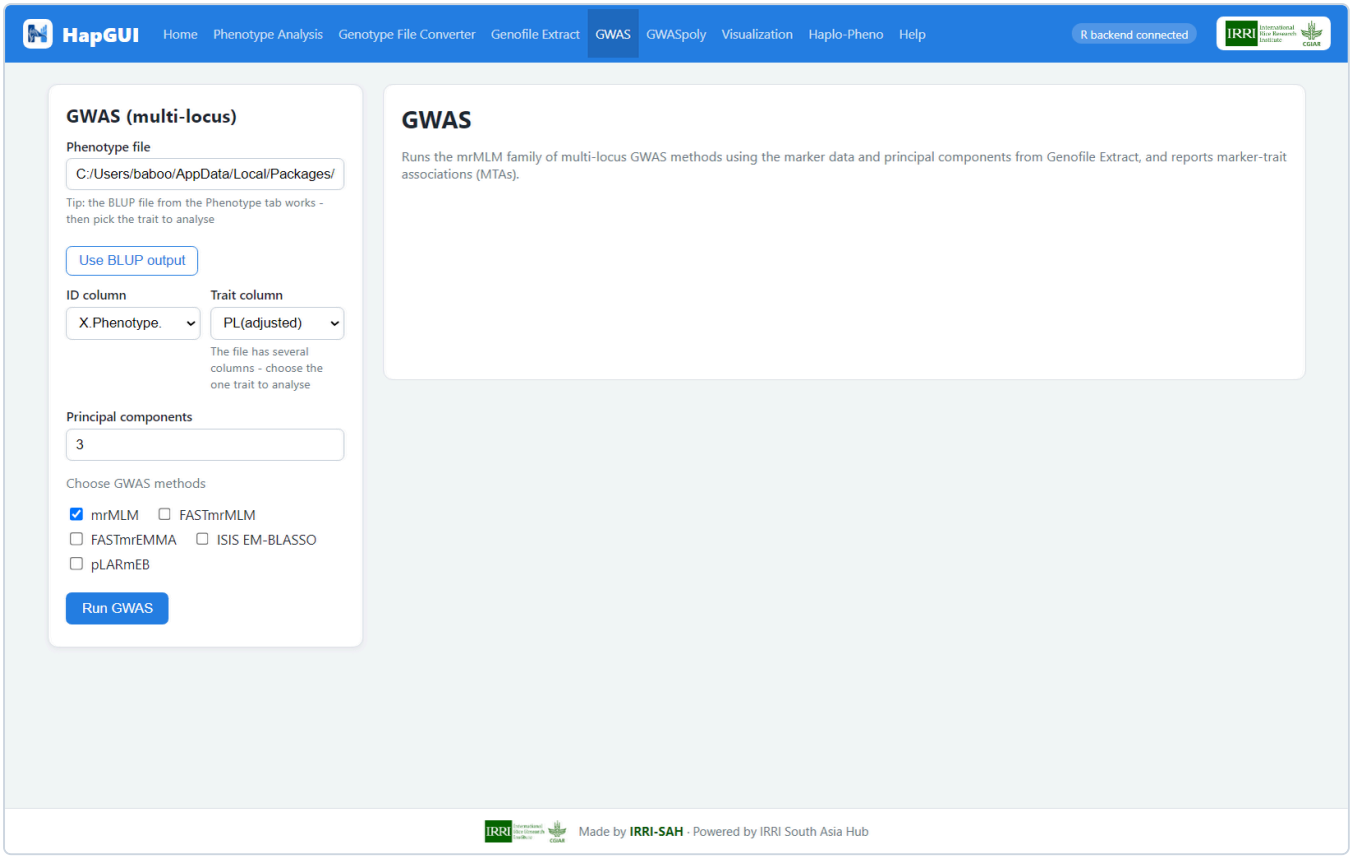


Figure 13. Choosing the phenotype file and the ID and trait columns (here the BLUP output with two traits).

> GWAS first checks that your phenotype lines are present in the genotype data from the last Genofile Extract. If none are, it stops at once and tells you which IDs it found on each side, so you can run Genofile Extract again with the same phenotype file.

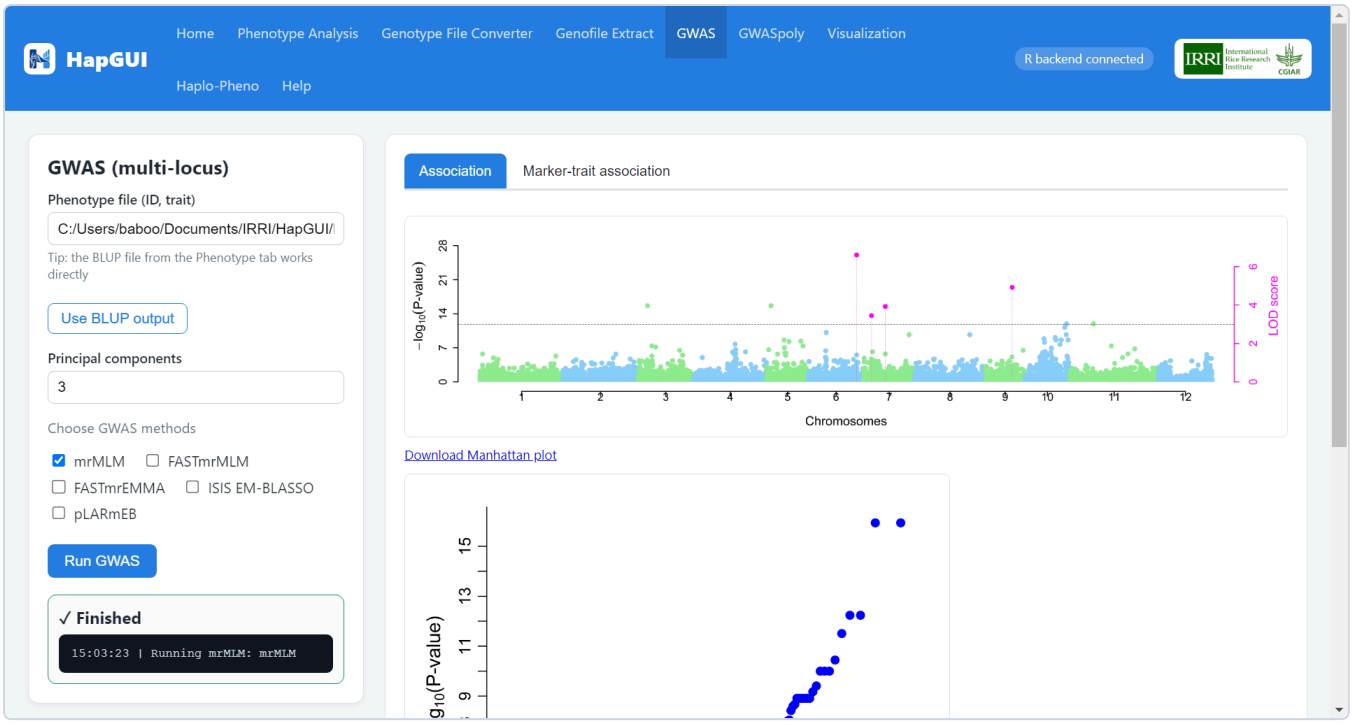


Figure 14. Manhattan plot (significant markers are highlighted) and QQ plot.

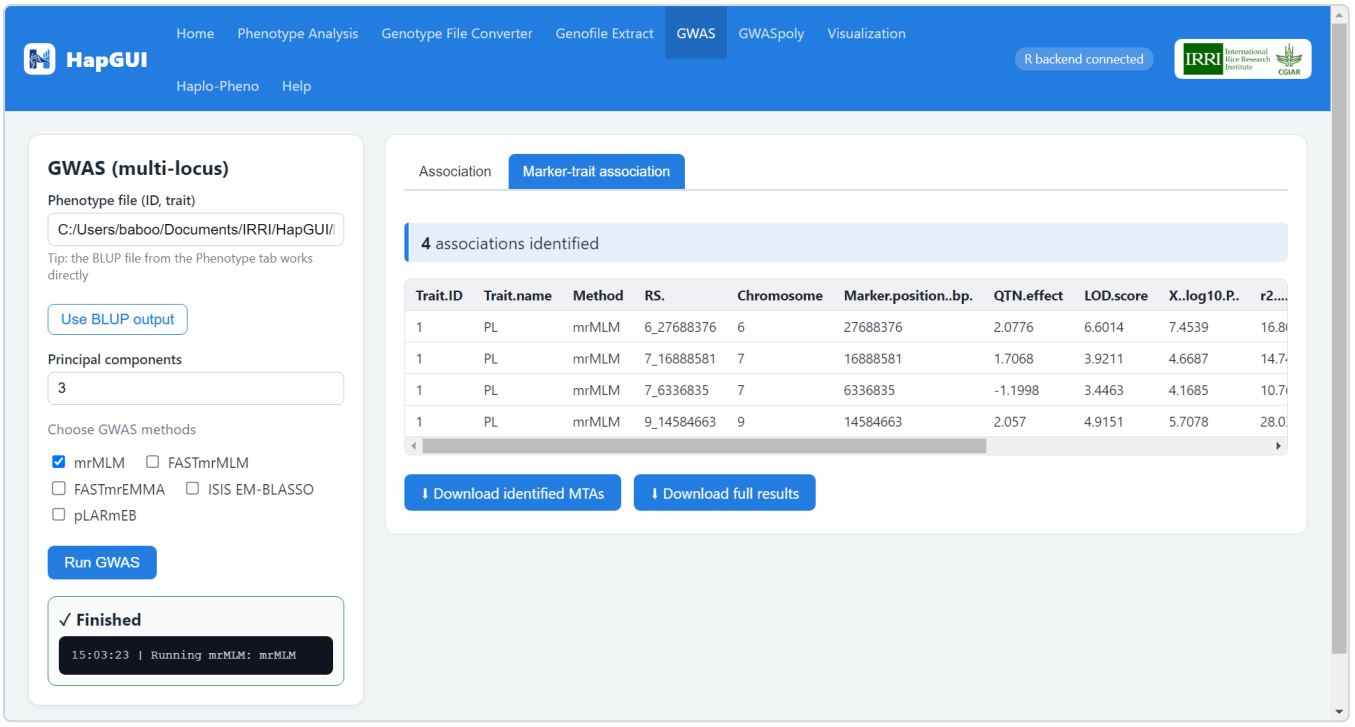


Figure 15. Marker-trait associations: chromosome, position, effect, LOD score, explained variance and the methods that detected it.

Significance threshold: LOD ≥ 3. If nothing passes, the tab reports that no associations were found. **Download identified MTAs** saves the file you can feed to Haplo-Pheno.

10. GWASpoly

Association analysis for **polyploid** species (and also usable for diploids), using the general, additive and simplex-dominant models with PC1–3 as covariates. Like GWAS, it needs Genofile Extract to have been run first.

Setting	Meaning
Phenotype file, ID column, Trait column	CSV with line IDs and a numeric trait; choose the two columns. The trait column name is used in file and plot names.
Ploidy level	Number of chromosome sets (for example 2, 4, 6).

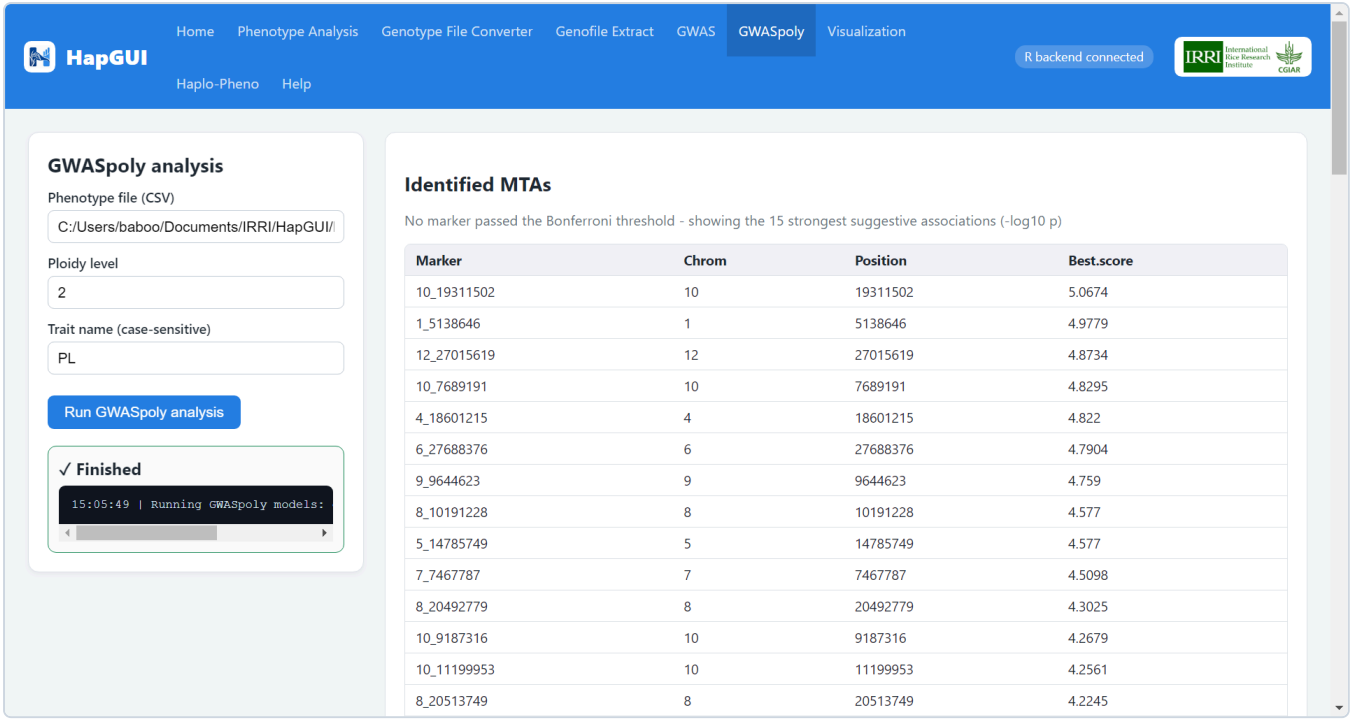


Figure 16. GWASpoly results: associations, Manhattan plots and QQ plots per model, and the LD plot.

Associations are reported at the Bonferroni threshold (α = 0.05). If no marker passes it, the 15 strongest suggestive associations are shown instead, and the note above the table says so.

11. Visualization

Draws publication-style figures from your own result files. Choose a tab, select the file and click **Run**.

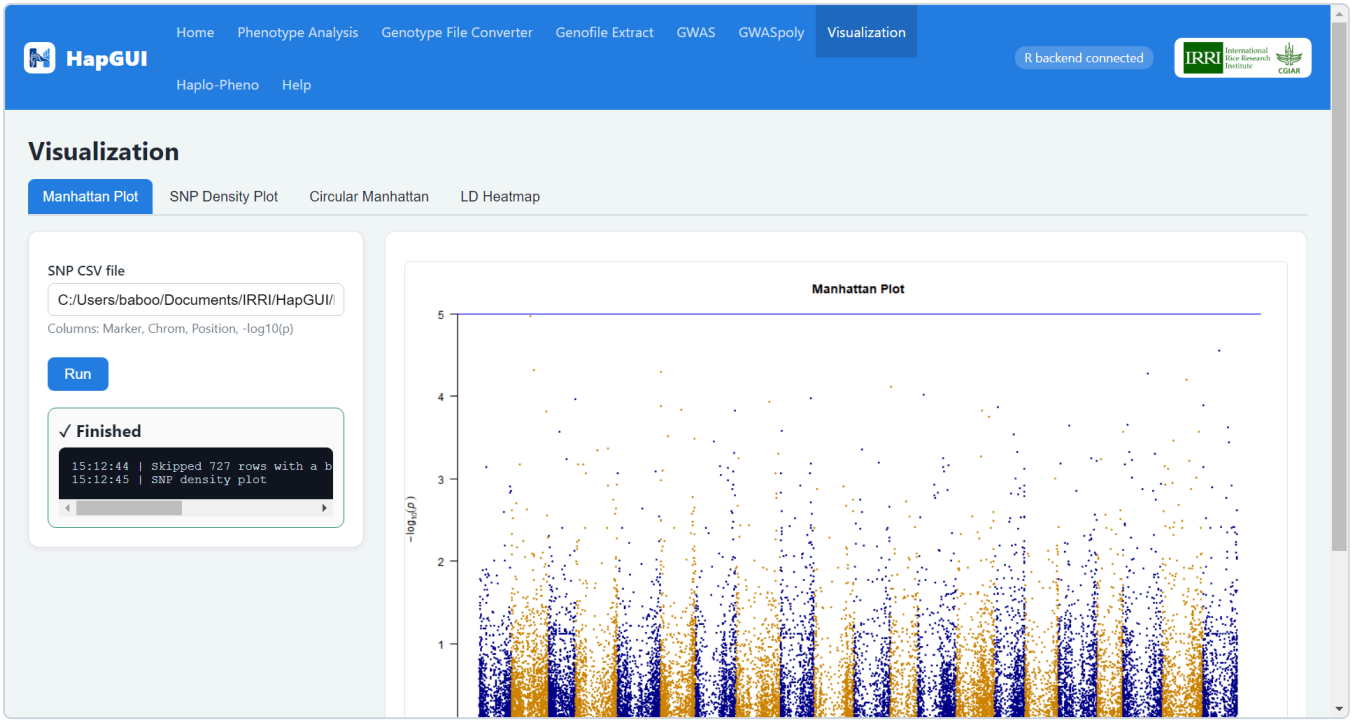


Figure 17. Manhattan plot. The SNP Density Plot tab shows the marker density from the same file.

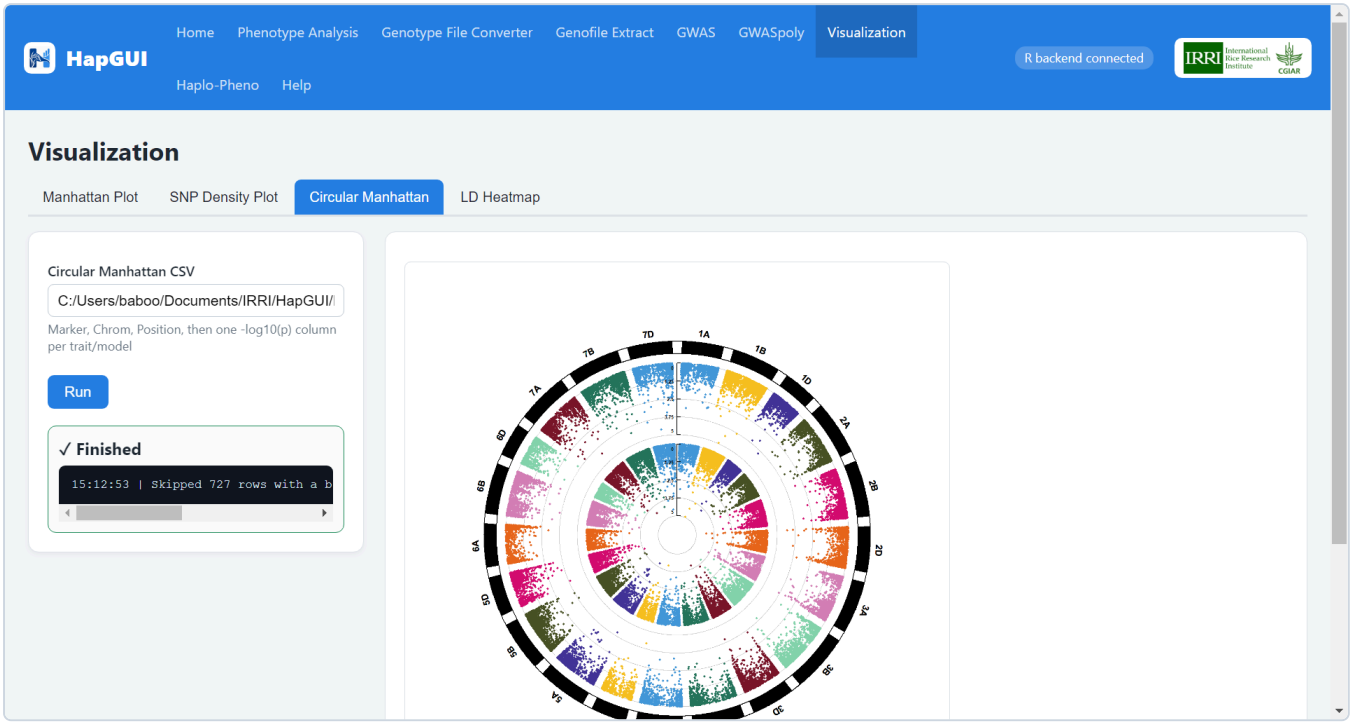


Figure 18. Circular Manhattan plot.

The **LD Heatmap** tab takes a VCF and shows one heatmap per chromosome, each on its own sub-tab.

> Rows with a blank chromosome or non-numeric position/value are skipped, and the progress box tells you how many. If the fourth column holds raw p-values rather than −log10(p), convert it first.

12. Haplo-Pheno

Finds candidate genes around your marker-trait associations, builds haplotypes from the SNPs inside each gene, tests whether haplotypes differ in the trait, and lists the **superior haplotypes** with the **donor accessions** that carry them.

Setting	Meaning
Phenotype file, ID column, Trait column	Line IDs and the one trait to analyse (or **Use BLUP output**, then pick the trait).
MTA file	Chromosome and position of each association (or **Use GWAS output**).
LD region	Genes within this many base pairs on either side of each marker are candidates (default 25,000; range 1,000–250,000).
High / Low	Whether the superior haplotype is the group with the highest or the lowest trait values.
Max missing data per SNP (%)	SNPs in a gene with more missing calls than this (default 20 %) are left out of that gene's haplotypes. Lines that still have a missing call at the remaining SNPs are dropped.

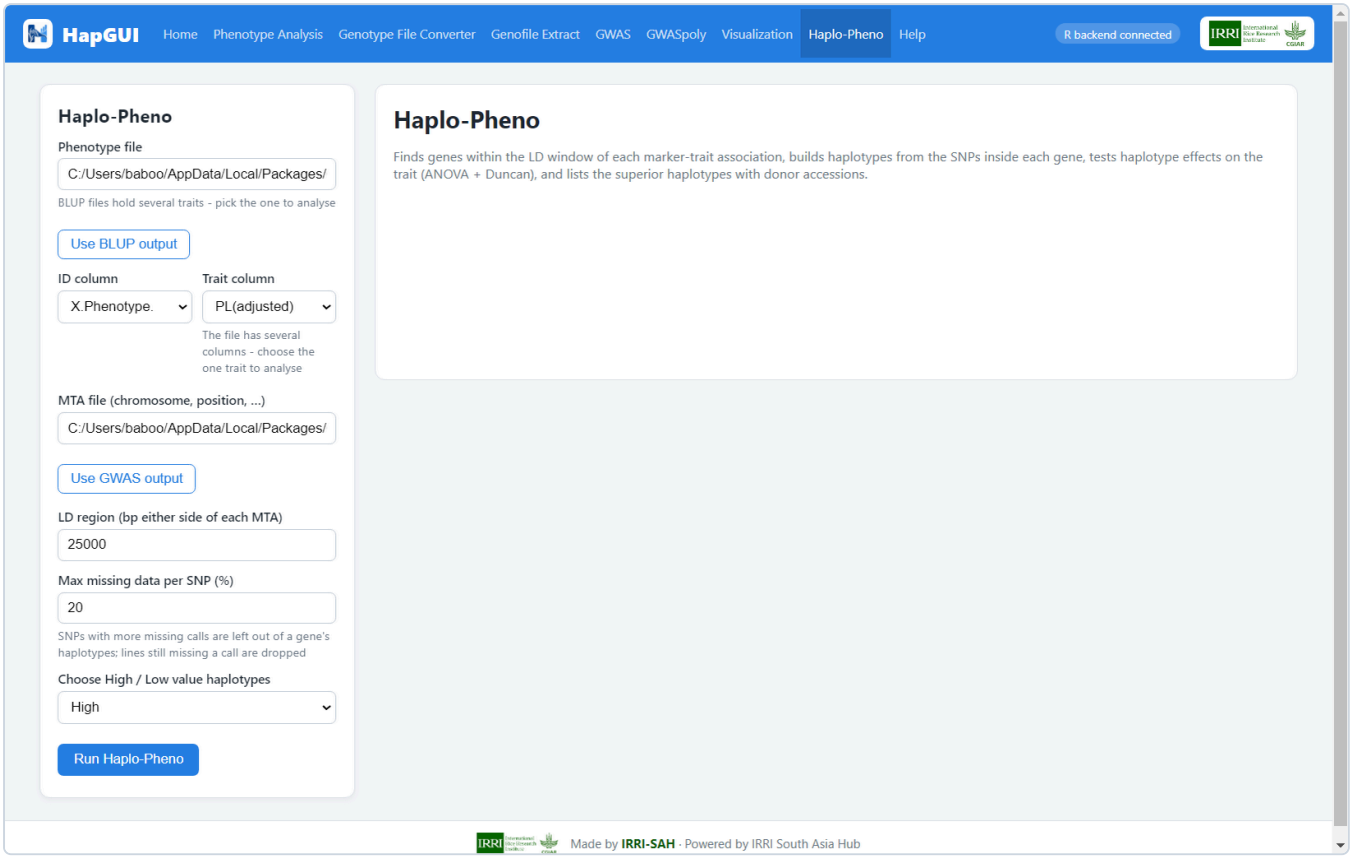


Figure 19. Haplo-Pheno settings.

Results

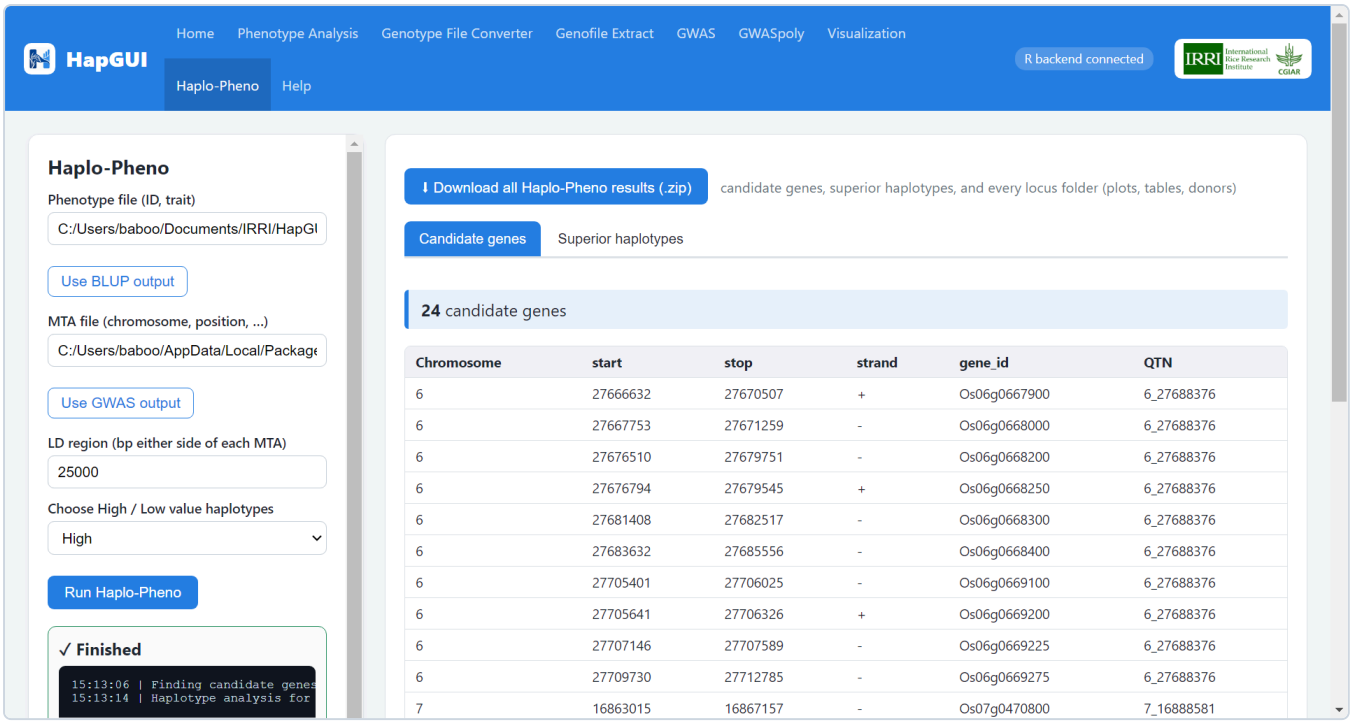


Figure 20. Candidate genes and the associated marker.

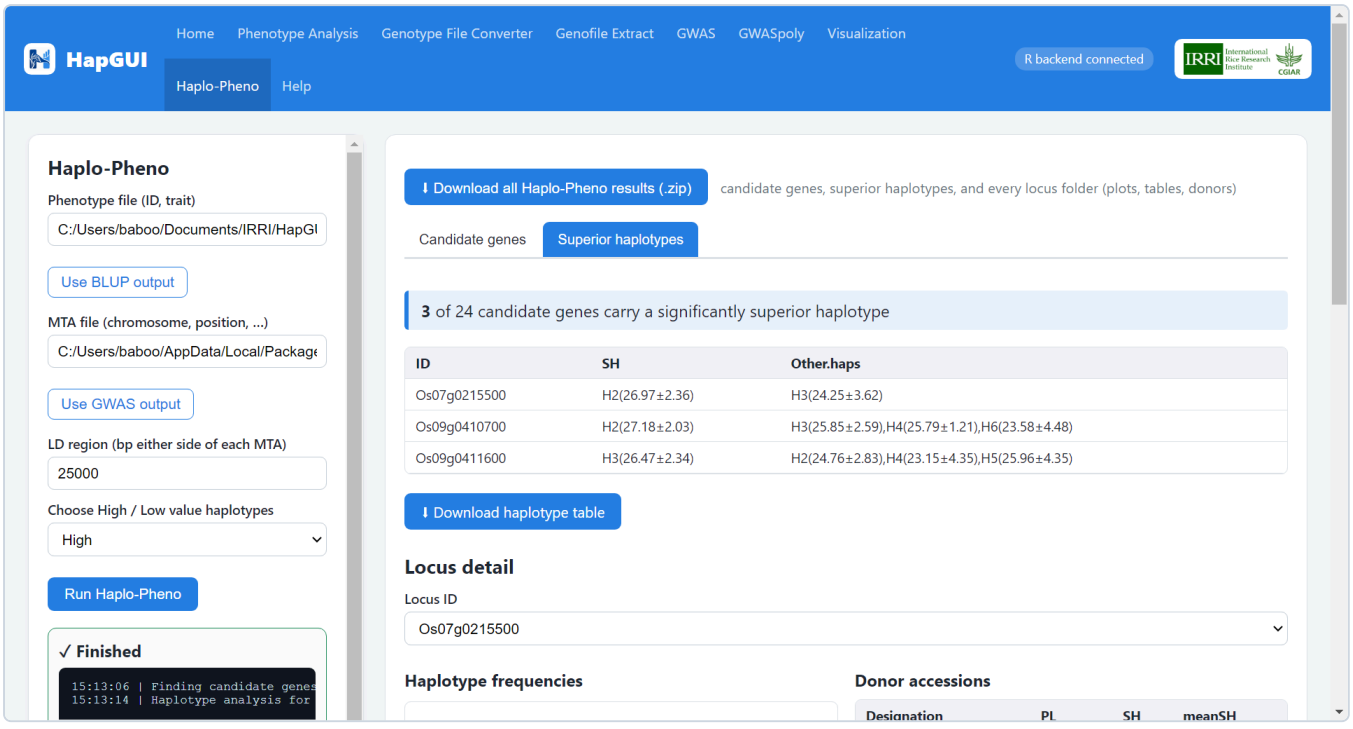


Figure 21. Genes with a significantly superior haplotype (the SH column) and the other haplotypes, as mean ± SD.

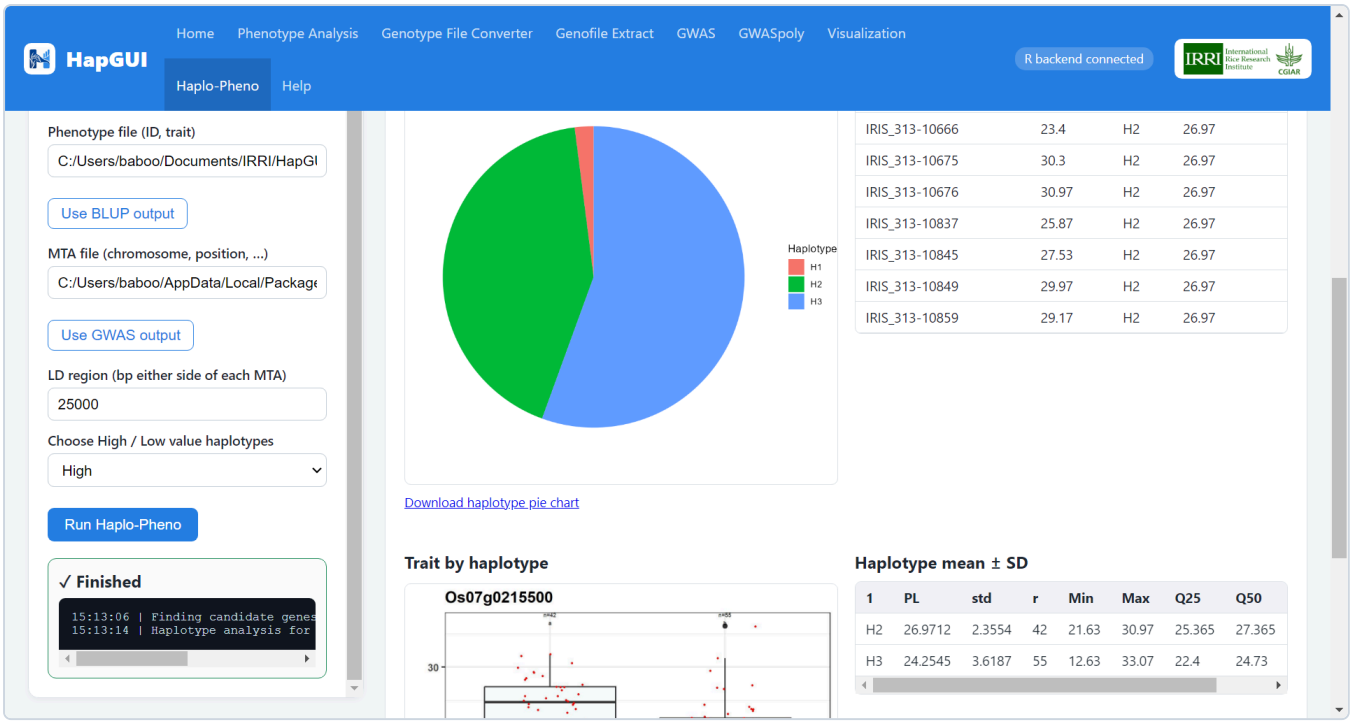


Figure 22. Locus detail: haplotype frequencies, donor accessions, trait by haplotype and haplotype means.

> **Download everything at once.** Above the result tabs, **Download all Haplo-Pheno results (.zip)** saves one zip containing the candidate-gene table, the superior-haplotype table and a folder for every candidate gene with its haplotype table, plots, donor list and statistics.

Markers that have no gene within the LD region are simply skipped. If *none* of your markers has a gene nearby, the tab says that no candidate genes were found; increase the LD region in that case.

How the haplotypes are called

- Only homozygous, non-missing calls are used. First, SNPs with too many missing calls (see **Max missing data per SNP**) are removed from the gene; then lines that still have a heterozygous or missing call are left out. A missing call is never treated as an allele.
- Identical SNP strings form a haplotype (H1, H2, …). Haplotypes carried by fewer than 3 lines are not tested.
- With two or more testable haplotypes, an ANOVA and a Duncan multiple-range test group the haplotypes. The top group (**High**) or bottom group (**Low**) gives the superior haplotype(s).
- If there is only one group, the gene is marked “no” (no significant difference). “NA” means no haplotype test was possible: the gene contains no markers in your genotype data (the results page says how many genes this affects), or no haplotype is carried by at least 3 lines. Sparse marker sets and many SNPs per gene both make “NA” more likely.

13. Where your results are stored

Results are written to a working folder, and every download link in the app saves a copy wherever you choose:

`%APPDATA%\hapgui-desktop\data\workspace` (paste this into the Windows Explorer address bar)

File / folder	Created by
`pheno.csv`, `blups.csv`, `summary*.csv`	Phenotype Analysis
`marker.csv`, `pca.csv`, `chrom_map.csv` (only if your chromosome names are not plain numbers), `annotated_variants.csv`, `marker_LD_stats.csv`, `snpsift\` (per-gene SNP files)	Genofile Extract
`GWAS_results\` (plots, `Identified_MTAs.csv`)	GWAS
`GWASpoly\<trait>\`	GWASpoly
`Haplopheno\`, `candidate_genes.csv`	Haplo-Pheno
`viz\`, `viz_ld\`, `converted\`	Visualization, Converter

> **New runs overwrite old ones.** Running Genofile Extract again replaces the marker, PCA and per-gene files; running GWAS again replaces `GWAS_results`. Download or copy anything you want to keep first. Results shown on screen are cleared when you close the app, but the files above stay on disk.

14. Troubleshooting

Problem	What to do
"R backend offline" (red badge)	The analysis engine stopped. The app restarts it automatically; if it stays red click **Restart** next to the badge. **Log** opens the log file (`%APPDATA%\hapgui-desktop\logs\backend.log`) – send it to us if the problem repeats. A job that was running when the engine stopped must be started again.
Extract stops or the engine goes offline on a big dataset	Almost always memory. Close other programs, and for very large VCFs filter the file to the lines and markers you need first (for example with PLINK) or run on a computer with more RAM.
"Run Genofile Extract first"	GWAS, GWASpoly and Haplo-Pheno need the files Extract creates. Run it first, with the same germplasm and phenotype lines.
"No samples from the phenotype file were found in the genotype data"	The IDs in the first column of your phenotype file do not match the genotype data. For 3K-RICE use IRIS IDs (`IRIS_313-…`); for external data use the exact sample names in the VCF.
GWAS finds no associations	Not an error: no marker reached LOD 3. Try more lines, a different trait, or another method.
Haplo-Pheno finds no candidate genes	Increase the **LD region**, and check that the chromosome names in the MTA file match the annotation (for 3K-RICE: 1–12).
"Something went wrong in this screen"	The screen showed an unexpected value. Your data is safe: click **Back to Home** and repeat the step. Please report the message shown (see Getting help).
Visualization says the file needs 4 columns or has no usable rows	Check the column order (Marker, Chrom, Position, −log10 p) and that Position and p are numbers.
Windows or antivirus warns about the installer	The installer is not code-signed. Choose **More info** → **Run anyway**. If your antivirus quarantines files in the install folder, add the HapGUI folder to its exclusions.
"None of the … phenotype lines … match a sample in the VCF"	The sample names in the VCF are spelled differently from the IDs in your phenotype file. Use the **ID mapping file** option in Genofile Extract (section 4.3) to rename the VCF samples.
Only some lines match (for example 79 of 186)	The VCF does not contain the other lines under any name; a mapping file cannot add them. The extraction continues with the matching lines. Compare the two ID lists, or use the built-in 3K-RICE data if your lines are 3K accessions.
Genofile Extract with a polyploid VCF	Supported (section 8). Use GWASpoly for the association analysis; mrMLM GWAS and Haplo-Pheno need diploid data.
"None of the … phenotype lines are in the genotype data from the last Genofile Extract"	The marker table was made from different lines than your phenotype file (for example 3K-RICE lines against an earlier external run). Check the germplasm choice on Home and run **Genofile Extract** again with the same phenotype file you give to GWAS.
"The trait column … has no numeric values"	You picked a text column (such as `subpopulation`) as the trait. Choose a numeric trait column.

“No candidate genes: the chromosome(s) in your MTA file … do not exist in the annotation”	The MTA file and the GFF3 refer to different chromosomes. Names such as 1 / chr1 / Chr01 are matched automatically, so this means the chromosomes really differ (for example different genome versions). Use an annotation of the same genome as your VCF.
Many genes show “NA” in Haplo-Pheno	Those genes contain no markers (or too few lines carry each haplotype). This reflects the genotype data, not an error. Use denser genotype data, or increase the LD region so genes with markers are included.
Correlation plot: “Too many columns to draw”	The environment / unit columns you chose produce more than 16 combinations. Choose fewer columns on the left (for example only Location and Season).
Page is blank or stuck	Close and reopen HapGUI. Completed results are on disk (section 13).

15. Software used

HapGUI bundles the following open-source tools; they are included in the installer and do not need to be installed separately.

Component	Used for
R 4.5	Statistics, plots and the analysis engine
PLINK 2	Selecting lines, allele frequencies, PCA, LD calculation
TASSEL 5 (Java)	Genotype format conversion and sorting
mrMLM	Multi-locus GWAS (mrMLM, FASTmrMLM, FASTmrEMMA, ISIS EM-BLASSO, pLARmEB)
GWASpoly	GWAS for polyploids
augmentedRCBD, lme4	ANOVA, BLUPs and genetic parameters
VariantAnnotation, GenomicFeatures, vcfR	Variant annotation against the gene models
agricolae, haplotypes	Duncan test and haplotype handling
ggplot2, psych, qqman, CMplot, LDheatmap	Figures
Electron, React	The desktop application and its interface

16. Getting help

Open the **Help** tab in the app for this manual and a link to report issues. Bugs and questions can be raised at `https://github.com/IRRI-South-Asia-Hub/HapGUI/issues`. When reporting a problem, please include: what you did (tab and settings), the message shown on screen, and the file `%APPDATA%\hapgui-desktop\logs\backend.log`.

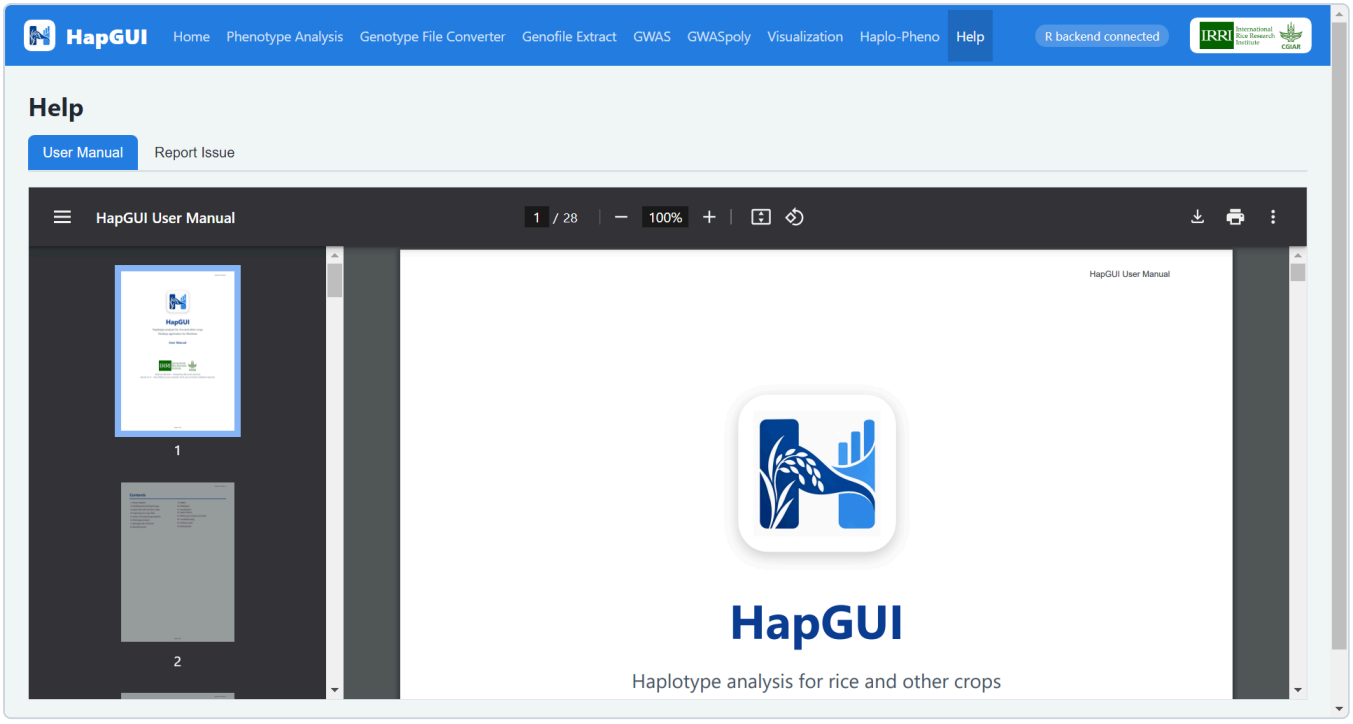


Figure 23. The Help tab shows this manual.